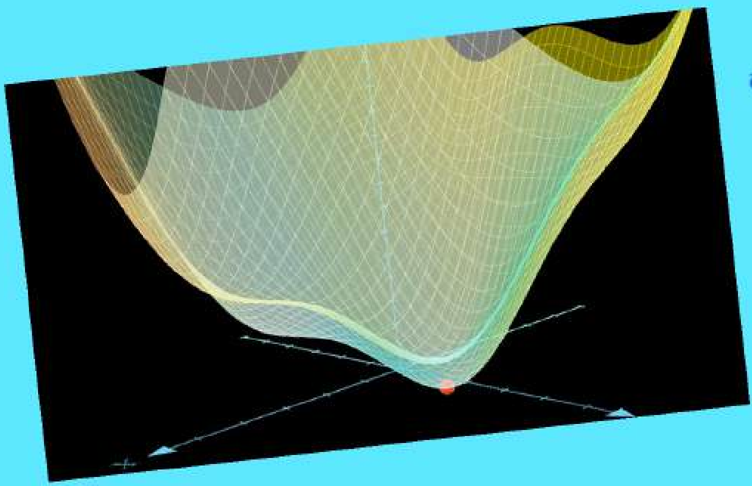

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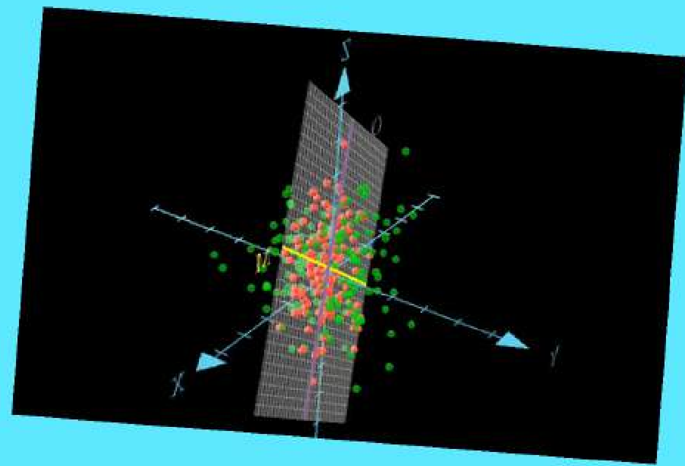
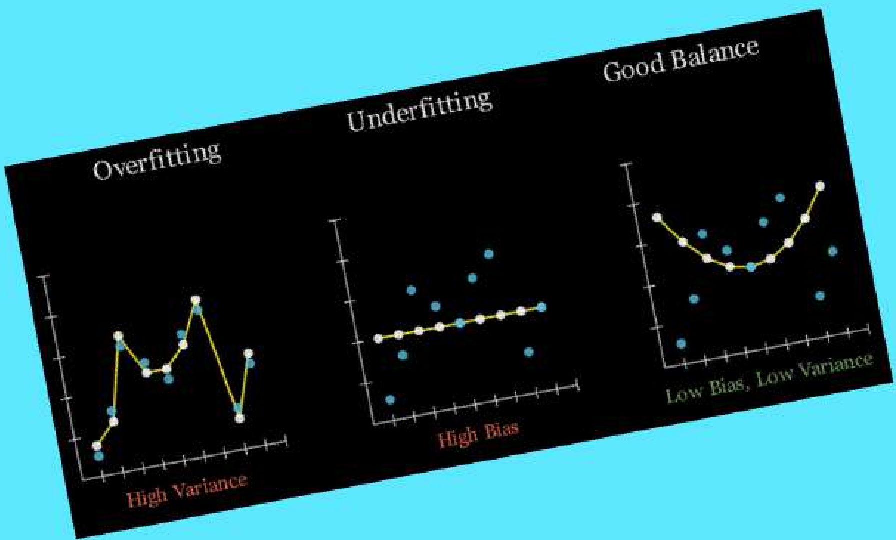
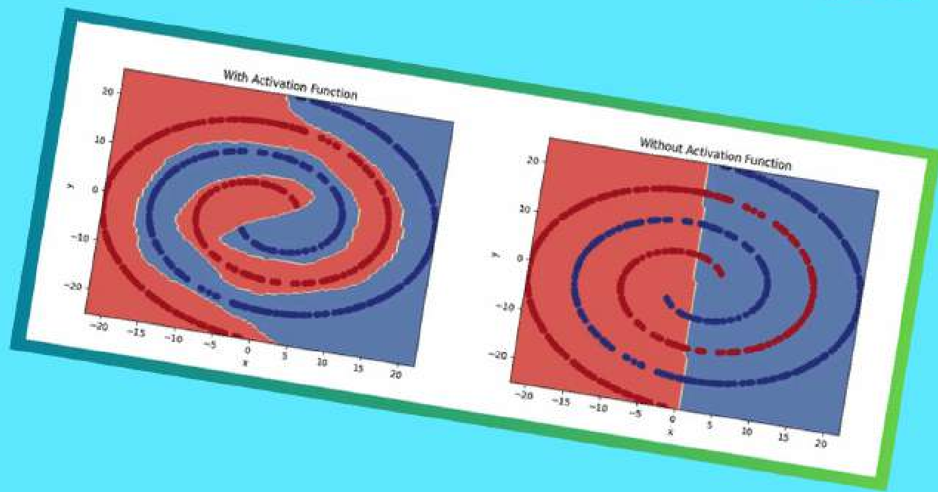






MOST ASKED

MACHINE LEARNING

Interview Questions

Rohit Kumar Tiwari

1. How do you explain the Bias-Variance tradeoff?

In machine learning, models face two fundamental types of errors that decide their performance.

> High Bias (Underfitting): When a model is too simple, it can't capture the underlying patterns of the data.
Example: Training accuracy is low (e.g. 65%) | Test accuracy is also low (e.g. 60%)

> High Variance (Overfitting): When a model is overly complex, it may learn not only the underlying patterns but also the noise in the training data.
Example: Training accuracy is very high (e.g. 97%) | Test accuracy drops significantly (e.g. 75%)

🎯 The Goal is to reduce bias without increasing variance too much, resulting in a model that performs well on both training data and unseen test data.

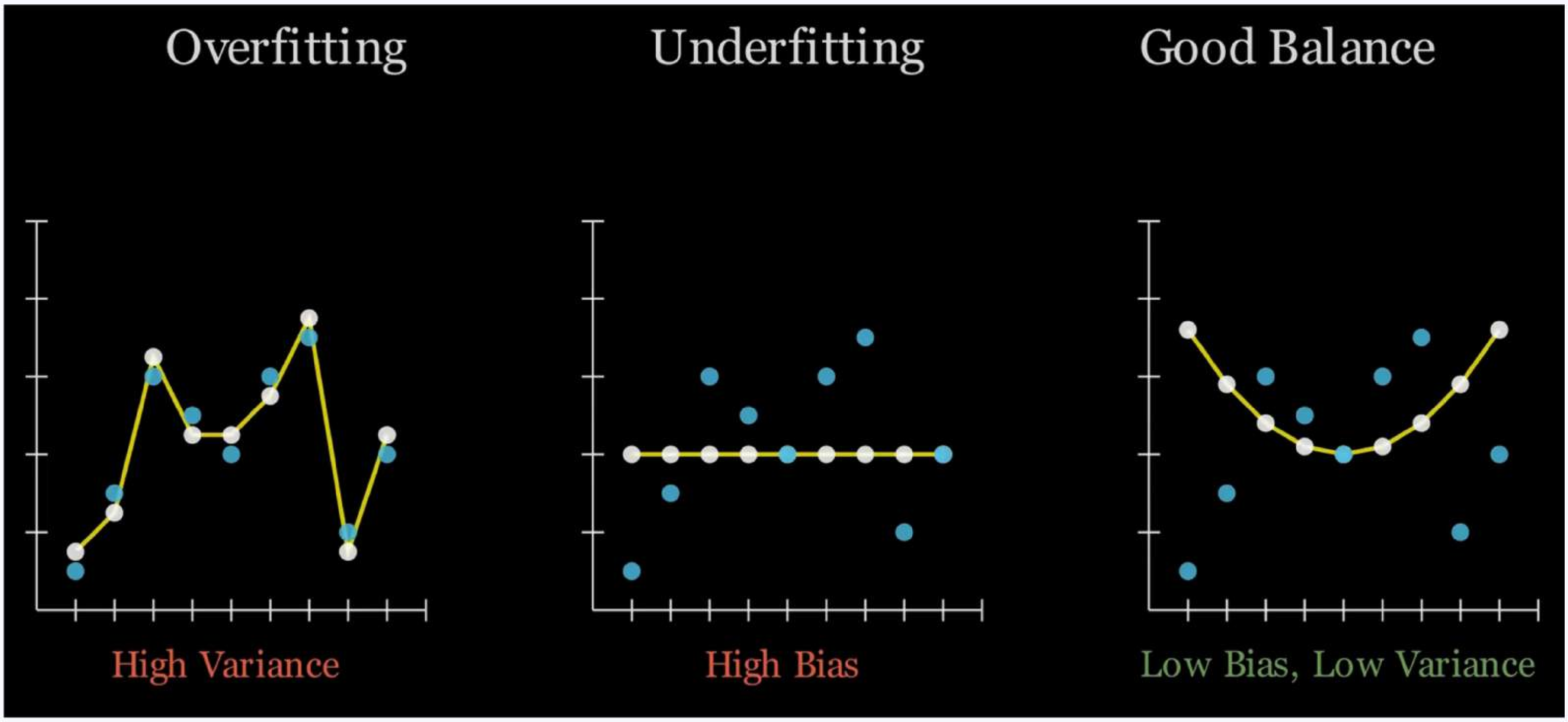
How to Solve This?

✅ Regularization: L1 (Lasso), L2 (Ridge), and Dropout can reduce overfitting.

✅ Feature Engineering: Keeping relevant features while excluding noisy ones.

✅ Ensemble Methods: Combine models (e.g., Random Forest, Gradient Boosting) to improve predictions.

✅ Hyperparameter Tuning: Performing hyperparameter tuning to avoid overfitting.



https://awesomeneuron.substack.com/p/bias-variance-tradeoff-in-machine

2. What is Gradient Descent in machine learning?

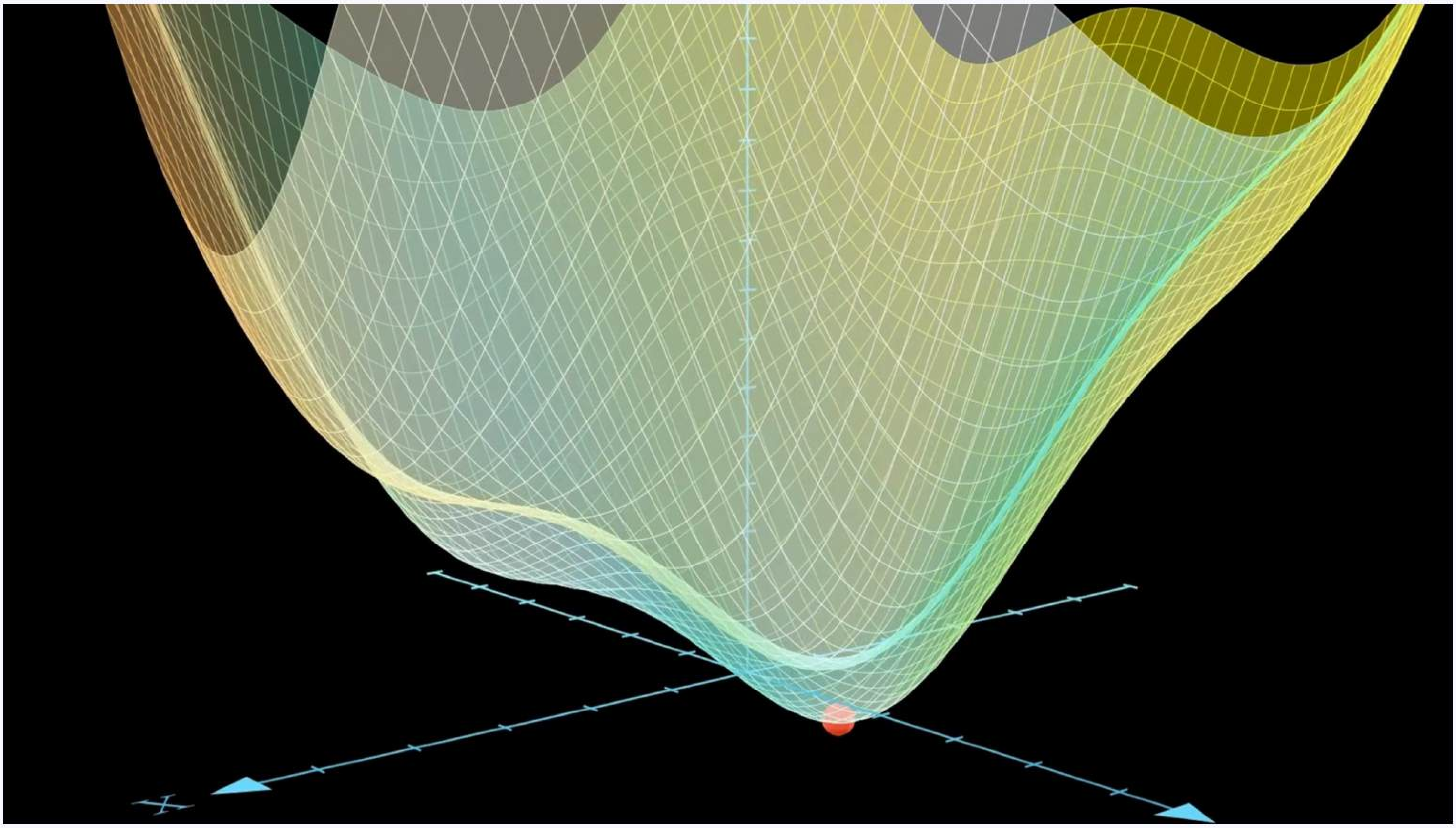
Gradient descent is at the core of most machine learning models. It’s the optimization technique that trains models and neural networks by fine tuning parameters to improve performance.

What is Gradient Descent?
> In simple terms, the "gradient" refers to the rate of change or slope of a function and "Descent" simply refers to going downhill.
> It is an algorithm used to minimize the cost function of a model. It adjusts model parameters (weights) iteratively to reduce the error between predicted and actual values, resulting in a more accurate model.

How it works?
Imagine you're trying to find the lowest point in a valley. To do so, you take small steps, each time adjusting your direction to follow the steepest slope.

✅ Gradient: The steeper the gradient, the bigger the step the model takes to minimize error.
✅ Learning Rate: This determines the size of each step. If it's too big, you might overshoot the minimum; if it's too small, convergence will take longer.
✅ Cost Function: The goal of gradient descent is to minimize the cost function until the model is as accurate as possible.

Types of Gradient Descent?
> Batch Gradient Descent: Sums the error for each point in a training set, updating the model only after all training examples have been evaluated.
> Stochastic Gradient Descent (SGD): Runs a training epoch for each example within the dataset and it updates each training example's parameters one at a time.
> Mini-Batch Gradient Descent: Combines concepts from both batch gradient descent and stochastic gradient descent. It splits the training dataset into small batch sizes and performs updates on each of those batches.



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3. What does dimensionality reduction mean, and how does PCA work?

PCA is a dimensionality reduction method. It reduces the number of variables or features in a data set while still preserving the most important information like major trends or patterns.

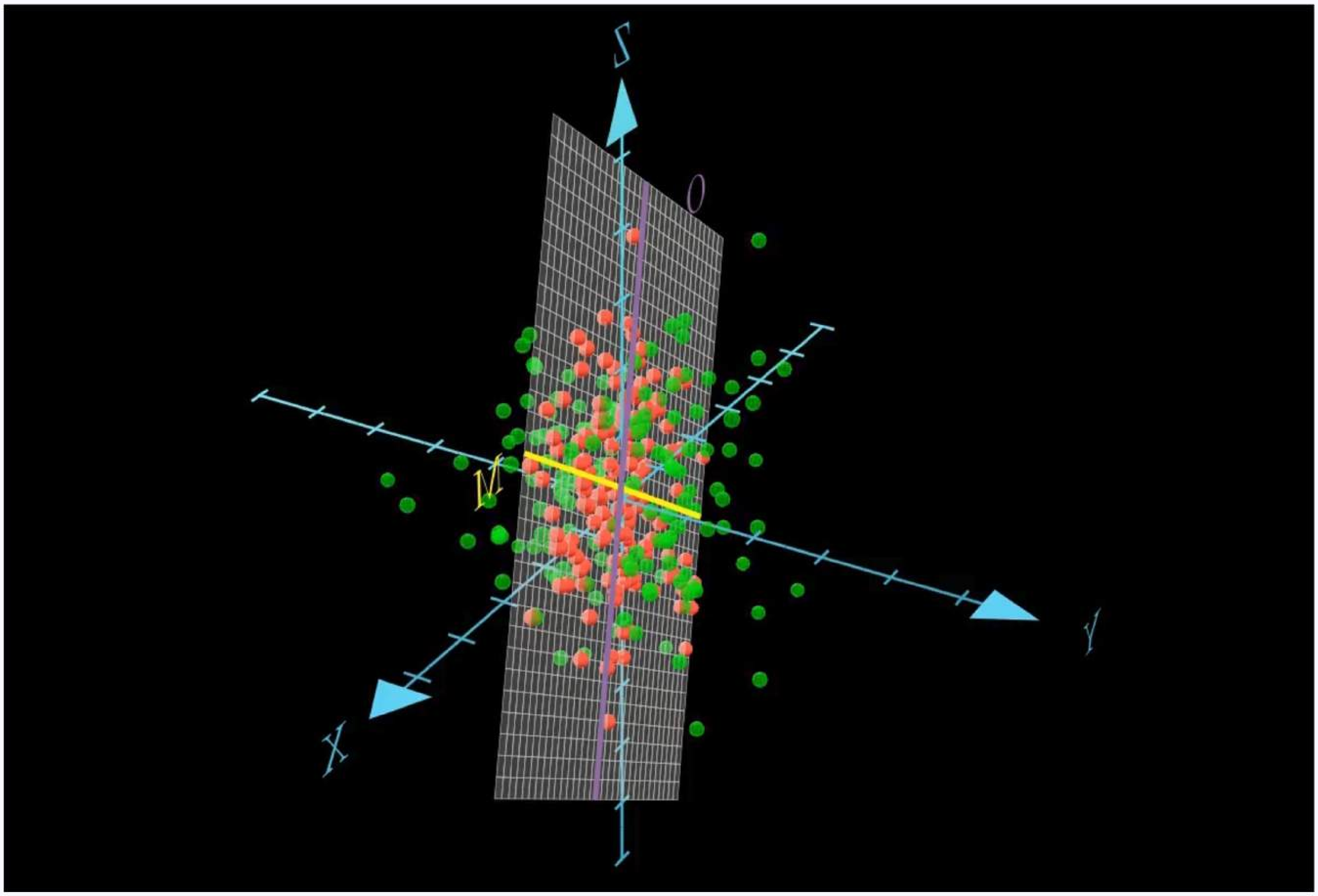
How does PCA work?
1. Standardization: Standardize your dataset to ensure all features are on the same scale.
2. Covariance Matrix Computation: Calculate the covariance matrix to measure how your features vary together.
3. Eigenvectors & Eigenvalues: Compute these from the covariance matrix to identify the principal components that capture the most variance.
4. Feature Vector Formation: Select the top principal components (based on their eigenvalues) to form a new, reduced feature set.
5. Data Projection: Transform your original dataset by projecting it onto the axes defined by your principal components.

Advantages of PCA?

✅ Data visualization: Easily visualize high-dimensional data by projecting it into 2D or 3D space.

✅ Multicollinearity Handling: PCA creates new, uncorrelated variables that help address issues when original features are highly correlated.

✅ Noise Removal: PCA helps in eliminating the components with low variance (often noise), this can improve model performance.



https://awesomeneuron.substack.com/p/pca-a-visual-journey-through-dimensionality

4. What is Bagging, and how does the Random Forest algorithm utilize it?

Random Forest is a popular machine learning algorithm that builds multiple decision trees and combines their outputs to arrive at one final prediction. It’s like asking a group of people for their opinion and then going with the majority vote.

How does it work?
Instead of relying on one decision tree (which can be biased or overfit), Random Forest builds many trees using different random samples of data and features. It then aggregates (majority vote for classification or averaging for regression) the results of these multiple decision trees to make a robust final prediction.

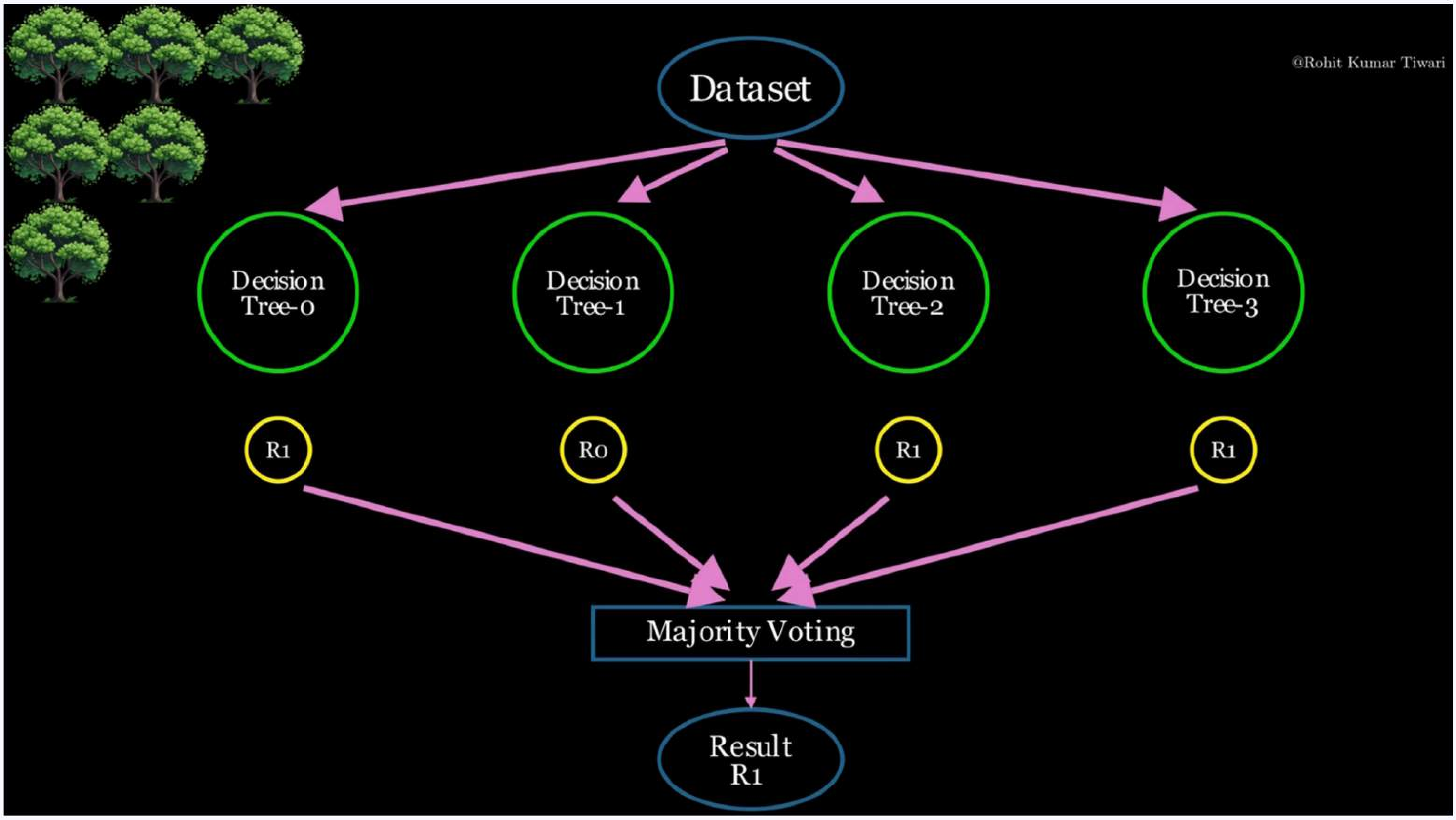
🎯 Whether you need to classify data (e.g. Yes/No for a credit application) or predict a continuous outcome (e.g. estimating house prices), Random Forest can handle both classification and regression tasks.

Why does it reduce overfitting?

✅ Random Forest employs bagging (bootstrap aggregation) and feature randomness.

✅ By training each tree on a random subset of data and a random subset of features, the algorithm minimizes correlations among trees.

✅ This ensemble approach significantly reduces overfitting.



https://awesomeneuron.substack.com/p/random-forest-algorithm-from-decision

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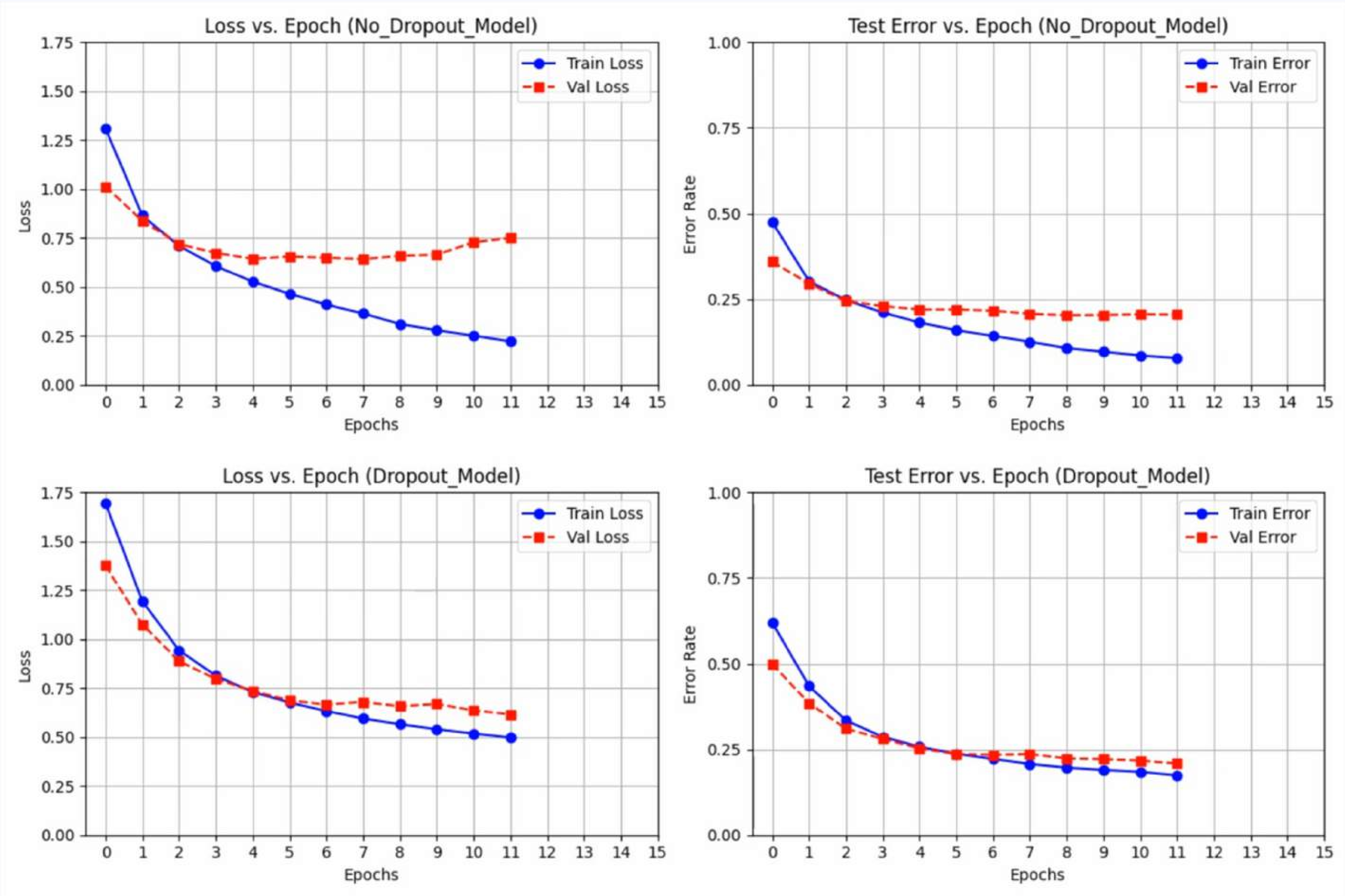
5. What is Regularization, and how does dropout technique help prevent overfitting?

When training deep neural networks, overfitting is a major problem. Your model may perform exceptionally well on training data but struggle on unseen test data. Dropout regularization is one technique used to tackle overfitting problems in deep learning. Dropout help prevent networks from memorizing the training data, forcing them to learn meaningful patterns.

How does Dropout work?
During training, dropout randomly disables (sets to zero) a fraction of neurons in each layer. This forces the network to learn more robust, distributed representations, making it more resilient and reducing overfitting.

✅ Prevents overfitting
✅ Improves generalization

Important point to note, during inference, dropout is turned off, and all neurons are used.



https://awesomeneuron.substack.com/p/dropout-regularization-a-simple-trick

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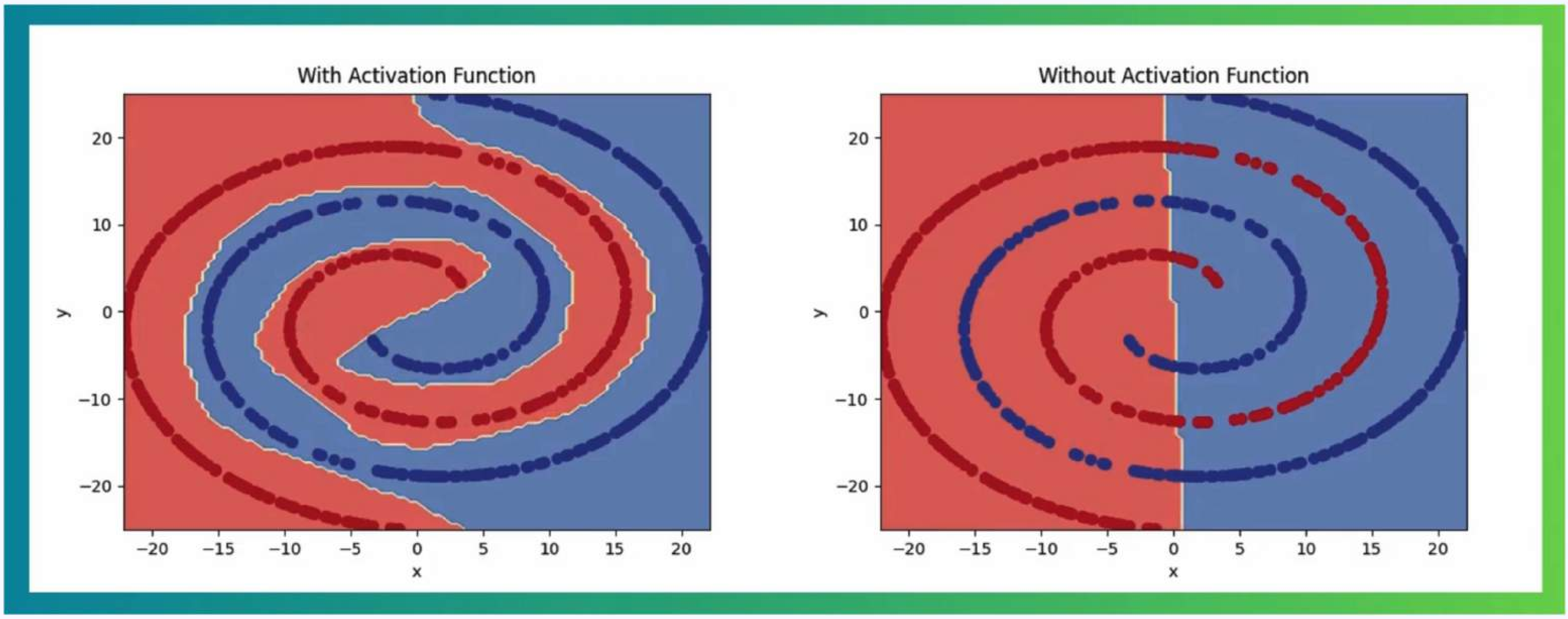
6. Why are Activation Functions important in neural networks?

Activation functions help neural networks learn complex patterns. They take the input from one layer, apply a mathematical function, and pass the result to the next layer. This function enables the network to model non-linear relationships.

Why are they Essential?
Without activation functions, neural networks would be restricted to modeling only linear relationships between inputs and outputs.

🎯 A neural network calculates the output of each neuron using the formula y = wx + b, where w is the weight, x is the input, and b is the bias. Even if you have 1,000 neurons or layers, the network would still only combine these calculations in a linear way. All layers would perform linear transformations of the input, and no non-linearities would be introduced.

✅ Activation Functions are foundational in models ranging from basic multilayer perceptron to advanced architectures like GPT, enabling these networks to capture complex, non-linear dependencies in data.



https://awesomeneuron.substack.com/p/activation-functions-the-secret-sauce

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7. What is a Z-score, and how does it help in detecting outliers?

Ever wondered how raw numbers can sometimes be misleading? Let me breakdown one of the most powerful statistical concept: Z-Score

👉 Meet Shivani. She secured:

✅ Physics: 85/100
✅ Maths: 80/100

At first glance, it seems she performed better in Physics. But when we standardize her marks, the story changes, Shivani actually performed better in Maths!

What is a Z-Score?
A Z-Score tells you how many standard deviations a particular value is away from the mean of its distribution. In simple terms, it standardizes data, making comparisons easier across different scales and distributions.

Why it matters?

Standardization: Transform various datasets into a common scale (mean = 0, standard deviation = 1) for direct comparison.

Outlier Detection: Identify values that are extremely high or low (typically beyond ±3) which might skew your analysis.

Real-World Applications

Education: Evaluate test scores relative to class performance.

Healthcare: Determine if a patient's reading is within normal ranges.

$$\text{Z-score} = (X-\mu)/\sigma$$

Where:

X = individual value

μ = mean of the dataset

σ = standard deviation

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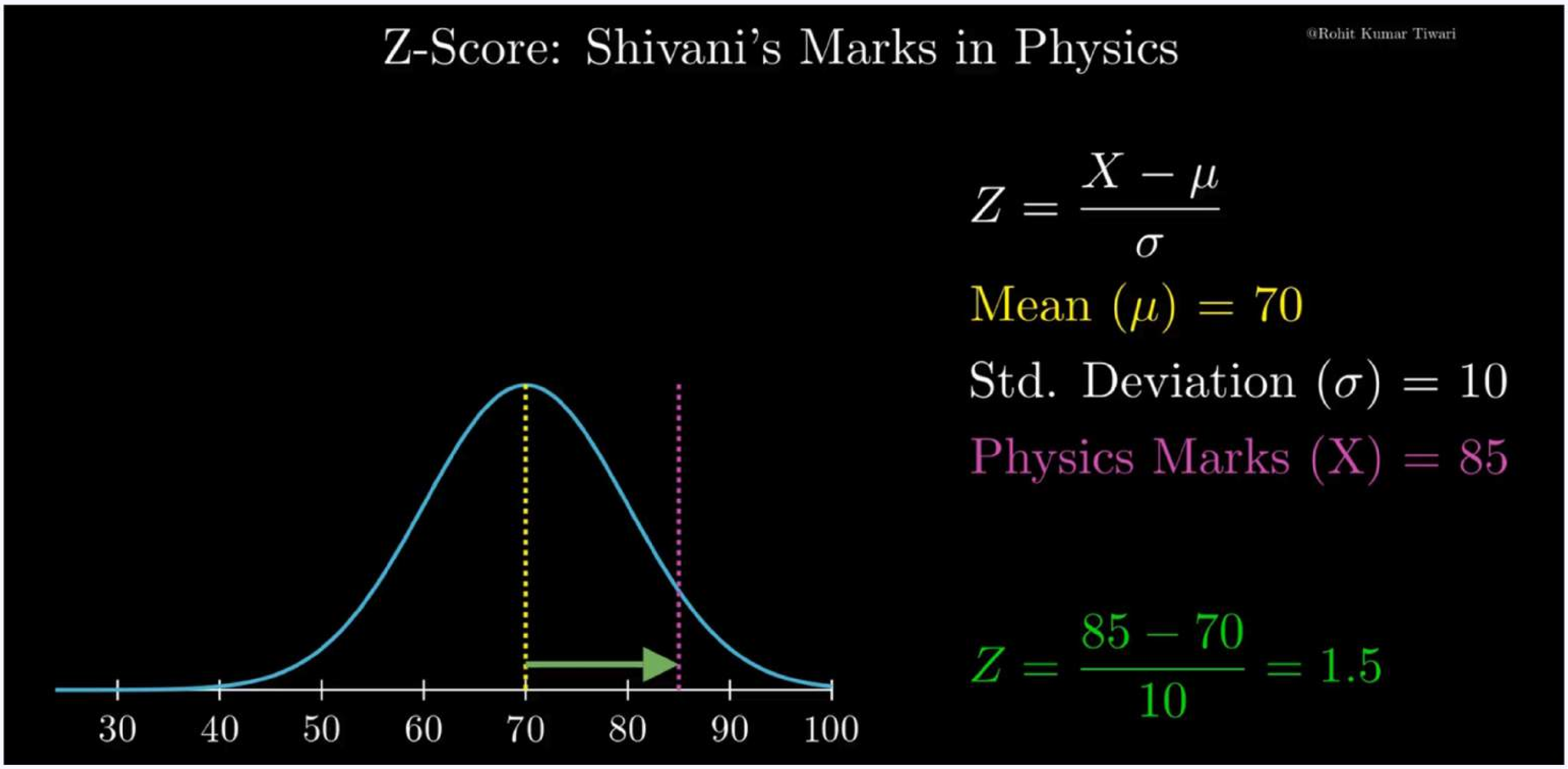

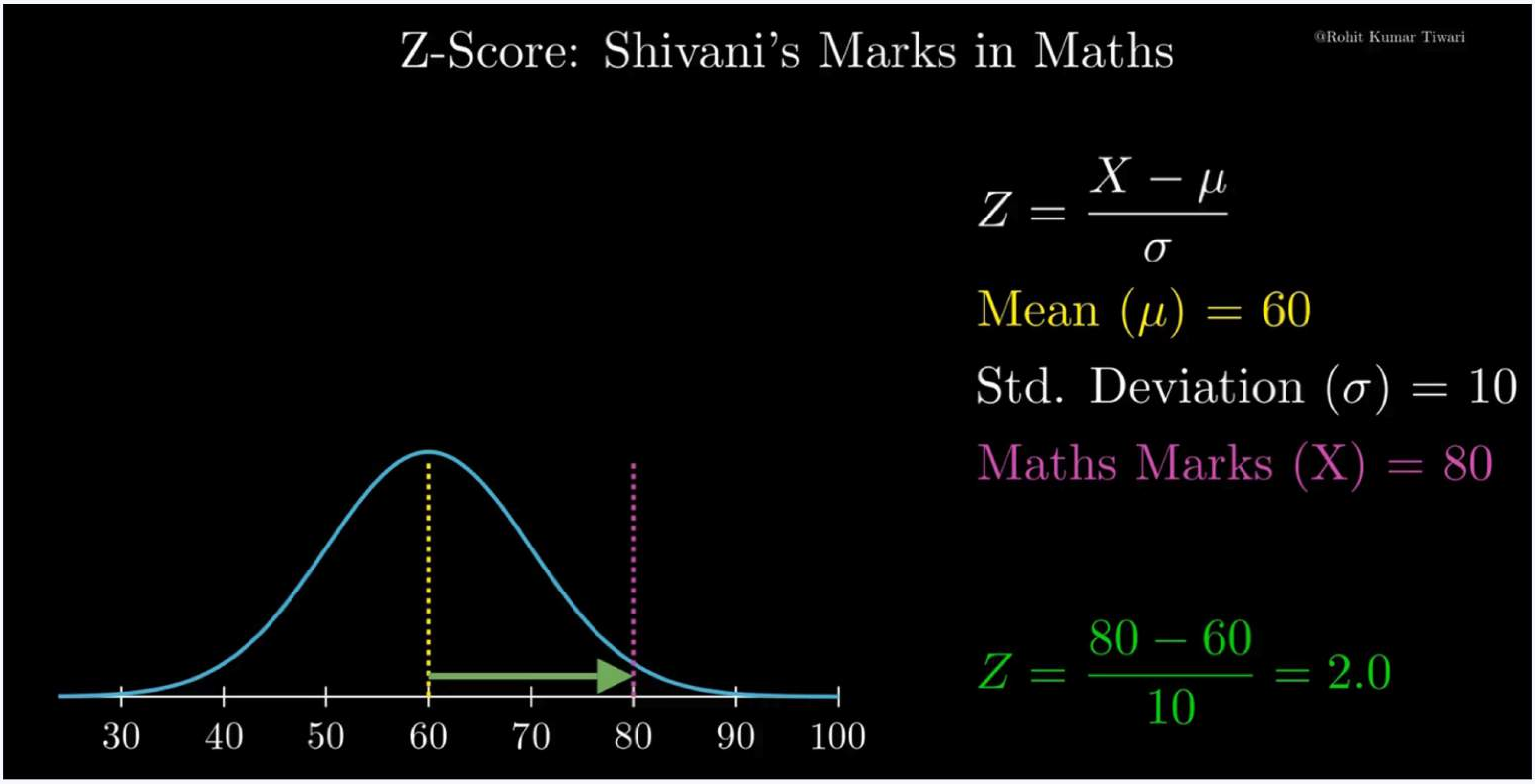

https://awesomeneuron.substack.com/p/beyond-the-numbers-how-z-scores-reveal

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Rohit Kumar Tiwari